[No of Questions : 03] [2612] – 2603MD16 Seat No 52602057

Society for Computer Technology and Research's

PUNE INSTITUTE OF COMPUTER TECHNOLOGY

INSEM EXAMINATION AUG-OCT 2026

S.Y. B.Tech (E&TCE) (SEM : III)

(2603MD16/0305161) (Data Structures and Algorithms)

Duration : One Hour Maximum Marks : 20

Instruction to the Candidates.

1. *Attempt Any Two Questions of Three Questions*
2. *Neat diagrams must be drawn wherever necessary.*
3. *[Marks], [Course Outcome] and [Bloom's Taxonomy Level] given at the end of each question.*
4. *Use of non-programmable calculators is allowed.*
5. *Assume suitable data if necessary.*

CO-1 : Recall and list variables, constants, keywords, data types, decision making techniques to perform data manipulation operations. Implement and analyze various sorting and searching algorithms, and evaluate their efficiency using asymptotic notations and real world datasets.

Q.1. A) Explain the concept of time complexity for O(log n) and O(n) with proper example [05 Marks] [CO1] [L2]

B) Write an executable C++ program that uses a recursive function to calculate the Fibonacci series up to the nth term, along with its time complexity. [05 Marks] [CO1] [L3]

Q.2. A) Explain working of Linear Search with example. Also write difference between Linear and Binary Search. [05 Marks] [CO1] [L2]

B) Write a C++ program to find out the occurrence of vowels and consonants from given input string.
Input string s = "MahaRaShTrA" [05 Marks] [CO1] [L3]

Q.3. A) Sort the following elements using Insertion Sort: [25, 10, 40, 15, 30] in Ascending order. Also write time complexity of Insertion sort. [05 Marks] [CO1] [L3]

B) Write a C++ program to remove duplicate entry from the sorted array.
Input Array is [1,2,2,3,3] [05 Marks] [CO1] [L4]

********** BEST OF LUCK **********